MARS MISSION DESIGN LAB / STUDENT PORTFOLIO

★ Safe Touchdown

Student / team: Student Engineering Team

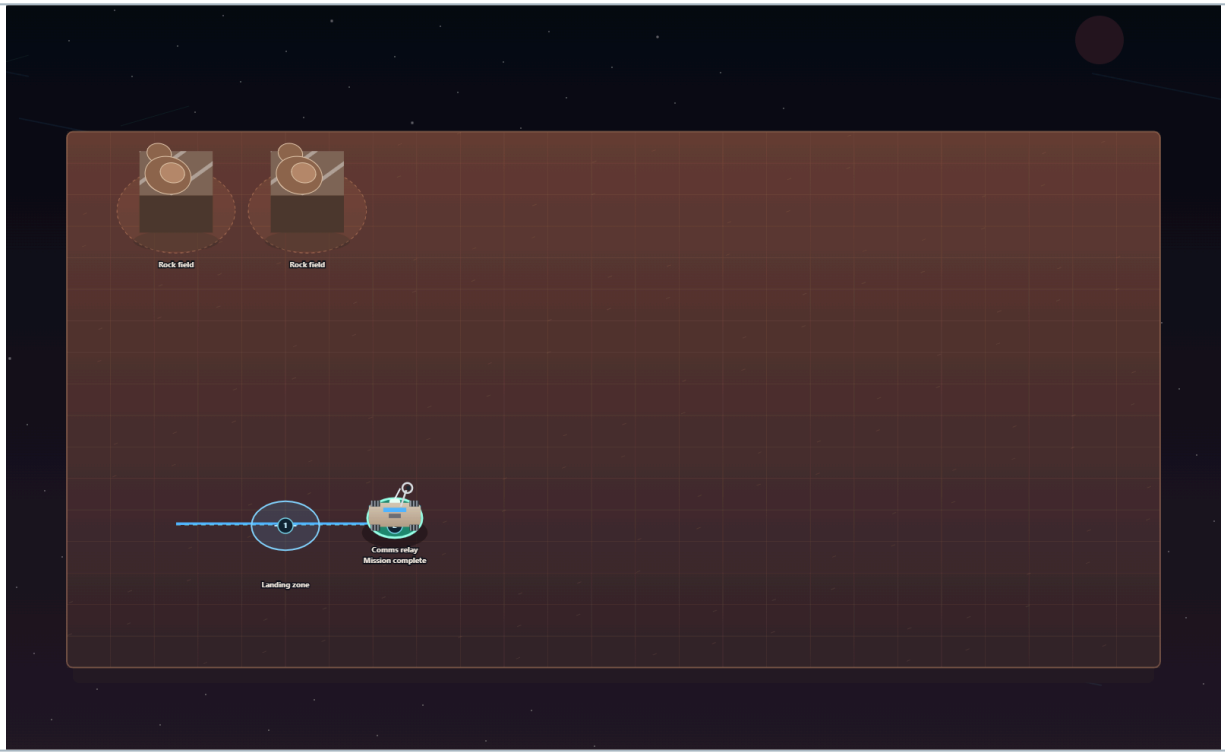
Mission challenge

Rank 1 · Mission 1/3 · Safe Touchdown
Navigate to a safe landing zone, call land(), then confirm touchdown at a relay. Include at least 2 obstacles to avoid.

Engineering rank

Flight Director · 2200 points · 12/12 missions · 4/4 ranks complete
This design passed its Python mission checks.

Mission design screenshot



Design captured for this proposal. Dashed line: planned route. Solid line: latest coded traverse.

Mission resources and testing

> Build credits: 510 / 1,000 | Terrain obstacles: 2 / 2 minimum
> Planned route: 200 map units | Planned energy: 13 / 100
> Latest test: Mission complete
> Actual code run: 200 map units | 14 / 100 energy | 5 commands

Our science question

How can our rover meet this mission objective safely?

Our engineering decisions

Not yet written.

Our Python rover program

Student-written program, including comments and indentation.

```
speed = 100
roll(speed, 90, 0.7812500)
land()
roll(speed, 90, 0.7812500)
transmit()
stop()
```

Python demonstration checks

PASS: Confirm safe touchdown with land()
PASS: Send orange-blue at a relay after touchdown
PASS: Include at least 2 terrain obstacles (2/2 placed)
PASS: Drive safely through the terrain
PASS: Meet 1,000-credit build and 100 actual code energy limits
PASS: Demonstrate a successful Python program

Placed equipment and obstacles

Object	Type	Position (X, Y)	Credits
Mars rover	Mars rover	100, 500	300
Rock field	Rock field	100, 100	0
Rock field	Rock field	220, 100	0
Landing zone	Landing zone	200, 500	80
Comms relay	Comms relay	300, 500	130

Planned flight path

1. 200, 500
2. 300, 500

Independent classroom simulation. Imagined Mars terrain and simplified build credits, energy and mission actions.